

Simulation of Starlink Based Synthetic Aperture Radar Methods

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Abstract—This paper discusses the simulation of a passive synthetic aperture radar system using Starlink satellites as illuminators of opportunity (IoO). It discusses the challenges of reconstructing the downlink signal drawing on experimental work by Humphreys et al. and Gomez-del Hoyo et al., presenting an adaptable tool for future inquiry. Finally, this simulated signal is propagated to point targets to generate SAR imagery providing a strong starting point for future inquiry.

I. INTRODUCTION AND MOTIVATION

PASSIVE radar systems trace their origins to the very beginning of radar itself when Sir Robert Watson-Watt used a BBC Empire transmitter to detect a bomber from 8 km away [1]. At that point, it was used due to a lack of dedicated transmitters but today that lack of a needed dedicated transmitter is an advantage passive system still carries over a traditional system. It allows the system to image without announcing its presence which is a characteristic necessary for covert operations.

Over 90 years on from Watson-Watt's first experiment, the electromagnetic space is much different from the shortwave signals from few broadcast towers spread across the country. Possible signals come from the phones in our pockets to satellites hundreds of kilometers away. The recent proliferation of LEO satellite clusters, namely the Starlink cluster, has provided a consistent and strong signal source over the majority of the globe, with the added benefit of a consistently moving source which allows for synthetic aperture modalities. The feasibility of such a system was discussed at length by Sayin et al. in 2019[2] and demonstrated experimentally by Gomez-del-Hoyo et al. in 2024[3]. The purpose of this paper is to develop a suite of tools for the simulation of the passive synthetic aperture radar using a Starlink satellite as an illuminator of opportunity (IoO), first by creating a facsimile of the satellite downlink signal before propagating that signal and processing the result.

II. SYSTEM SNR

A. Satellite Characteristics and Geometry

The Starlink satellite constellation currently consists of around 10,000 satellites in a low earth orbit (LEO) of roughly 500 km. They are designed to provide satellite internet to the areas where conventional methods such as fiber may not be an option. Because of their LEO position, they are significantly closer to receivers on earth than existing geostationary providers (HughesNet's Jupiter 3 at 35,000 km)[4]. Because of the decreased distance, the Starlink system has significantly lower propagation loss, which allows Starlink to operate at higher frequencies and a higher bandwidth. The downside is that the position of the satellite now varies in the sky, and constant coverage requires a large number of satellites to work in tandem. This problem has been solved by the addition of satellites to the constellation continuously since the first launch in 2019. Figure 1 shows that on a random day, nine Starlink satellites are visible over the city of Boston.

The downlink capability of particular interest be-

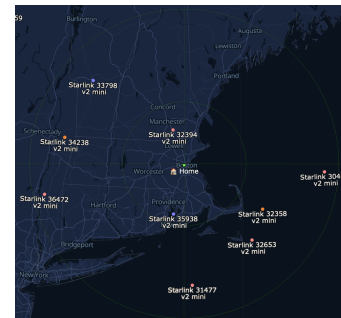


Fig. 1. Approximate positions of Starlink satellites around the city of Boston. This figure was generated using TLE data for Starlink Constellation on 4/5/2026 and the SatelliteMap.space visualization tool.[4]

cause its wide illumination capability. The satellites themselves each have a phased array antenna allowing the creation of seven individual beams that allow the illumination of several separate areas, known as

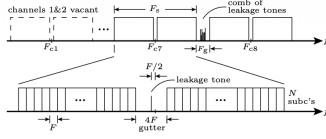


Fig. 2. Diagram of the eight channels of the Starlink downlink taken from Humphreys et al[5].

cells, simultaneously at separate frequencies. This also allows the 3 dB beamwidth of each beam to be kept at a maximum of 2.5 degrees[2] and with the altitude of 500 km yields an estimated spot size at nadir of 42 km.

Using the FCC filings from 2016, the characteristics of the satellites downlink transmitter can be ascertained. The transmitter has a maximum equivalent isotropic radiated power of 45.1 dBW with the phased array providing a peak gain of 34 dBi and operating in a range between 10.7 and 12.7 GHz[2].

B. Downlink Characteristics

Following the work of Humphreys et al.[5] who deconstructed Starlink downlink transmissions in 2023, the signal is divided into 8 separate transmission channels spanning the full allowed transmission bandwidth. Each channel spreads across 240 MHz, with the bandwidth being divided into 1024 subcarriers. Each channel has its own data stream which is divided into frames that are each 1/750 seconds long. Within each one of these frames are 302 separate symbols and an additional empty symbol that acts as a guard. The individual symbols vary across their position in the frame however several are unchanged from frame to frame. These unchanged symbols can be understood as synchronization sequences, with the first sequence being the primary synchronization sequence (PSS) and the second being the secondary synchronization sequence (SSS). Each frame begins with these two symbols. The PSS symbol consists of eight repetitions of a symmetric differential phase shift keyed sub-symbols broadcast across every subcarrier. The sub-symbol is an inverted version of the others and is prefaced by a cyclic prefix. The PSS sequence was observed to not only be consistent across frames but across different satellites. The values encoded in the PSS sequence is what's known as an m-sequence which is pseudo-random and provides it with excellent autocorrelation properties[5].

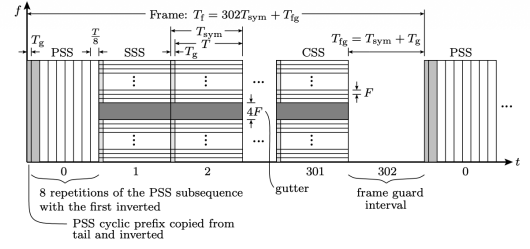


Fig. 3. Diagram of the symbol by symbol makeup of the downlink signal. Taken from Humphreys et al. [5]

The SSS sequence is composed of an entirely different makeup. It uses each of the 1024 subcarriers (with a 4 channel gutter around the center frequency) separately using orthogonal frequency division multiplexing (OFDM). Data is encoded on the subcarrier using quadrature amplitude modulation (QAM), specifically 4QAM, before being multiplexed. In alignment with OFDM methods a cyclic prefix is prepended onto the symbol to act as a guard and limit intersignal interference. The values encoded are thought to be scrambled combination of two m-sequences similarly to synchronization sequences in LTE signal[5]. Its value was found to not vary across satellites similarly to the PSS sequence. These sequences use the autocorrelation properties to synchronize the frequency and time between the satellites and the user terminals.

Humphrey et al. supposes that the next three frames are a header containing satellite position and system information due to parts of the signal being consistent across many frames. This is 4QAM encoded before being OFDM mixed across all 1024 sub-channels. The modulation constellation is then rotated 90 degrees and the payload section begins. The first frames (up to 7) of the payload section are modulated using 16QAM with 90 degree rotated constellation compared to the header. The number of 16QAM frames seems to depend largely on the SNR of the receiver. Following this section, the modulation returns to 4QAM with the constellation shifted from the header.

The final two frames contain what Humphreys et al. calls the coda minus one synchronization sequence (C1MSS) and coda synchronization sequence (CSS). The CSS sequence is consistent between frames and the C1MSS sequence is partially consistent, both encoded using 4QAM across all subcarriers using OFDM. The exact value of

these sequences is the subject of future work by Humphrey et al.[5]

Whether or not a frame is sent depends of the need of the user terminal. An example of what this total transmission looks like can be seen in Figure 4, which is taken from the work of Gomez-del-Hoyo et al. The frame repetition frequency gives

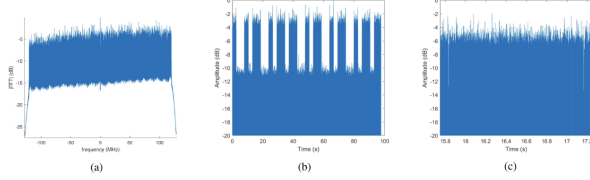


Fig. 4. Depiction of a recorded Starlink downlink signal taken from Gomez-del-Hoyo et al. (a) Normalized power spectrum after frequency drift correction. (b) Time domain of multiple frames. (c) Time domain of a single frame. [3]

us a pulse repetition frequency which we can use to calculate our bistatic maximum unambiguous range of roughly 400 kilometers. This is not the biggest limiter of our range resolution. The cyclic prefix of the OFDM symbols means that an autocorrelation will give sporadic peaks after the time difference of one symbol. This means that our maximum unambiguous range for the Starlink downlink frequency is 1320 m instead of the earlier 400 km estimate[3].

C. Link Budget Analysis

Given what we know about the Starlink system, we can now imagine a passive radar system that uses the downlink signal as an IoO, similar to that described by Blasquez-Garcia et al[6]. As a satellite passes over and illuminates the target area, a reference channel is recorded using phased array antenna tracking the and a signal channel is recorded by static directional antenna pointed towards the area in question. The differential range is then calculated from cross correlation of the signal and the reference. As the satellite travels

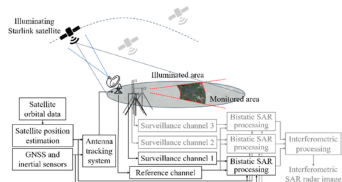


Fig. 5. Diagram showing the geometry and setup for a possible Starlink SAR setup. Reproduced from Blasquez-Garcia et al. [6]

across the sky, a synthetic aperture can be created through coherent addition of the pulses through backprojection.

Parameter	Value
Center Frequency	10.7 GHz
Bandwidth	240 MHz
Transmit Antenna Gain	34 dB
Receive Antenna Gain	20 dB
EIRP	45.1 dBW
R_{orbit}	500 km
RCS	100 m ²
T_n	290 k
Noise Factor	1.4 dB
Receiver Losses	6 dB

Using the number taken from the FCC filings[2] and the representative system design done by Sayin et al.[2], we can now calculate an expected SNR using the radar equation. For this situation, we can use

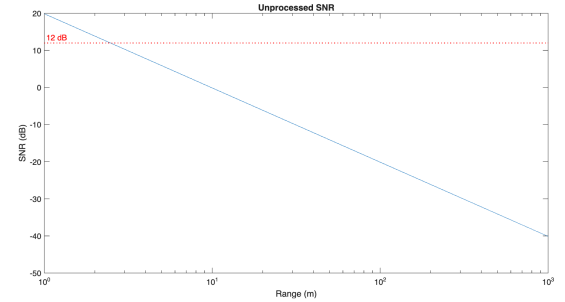


Fig. 6. Signal to noise as function of distance without added processing gain

12 dB as a detectability threshold and in Figure 6 we find that for our unprocessed signal, we would get a detectable signal at around 20 cm. This is not a promising result however, this model fails to account for the gain due to the matched filter and dwell time.

The compression gain (gain from pulse compression) of a system is equivalent to the time-bandwidth product of the pulse. We find that for one pulse, the compression gain comes out to $\frac{1}{750} s * 240 MHz = 320,000$ or 55 dB.

We can also use the number of pulses spent on the target to calculate our gain due to dwell. We know that the spot size on the ground at nadir is 42 km and that the satellite is traveling at 7.62 km/s. From this, we can determine that a target will remain in the spot for roughly 5.51 s or roughly 4133 pulses. A dwell of 4133 pulses comes out to a gain of 36

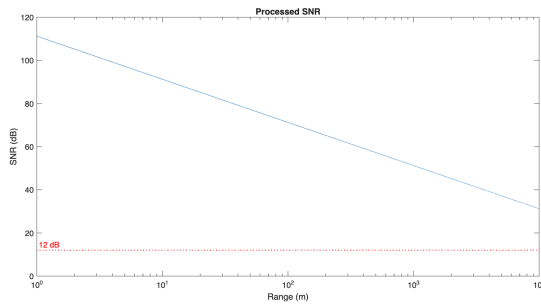


Fig. 7. Signal to noise as a function of distance after processing gain

dB. We take this additional gain into account when we recalculate our SNR. We can now see in Figure 7. that we are above our detectability threshold for well beyond our unambiguous range, echoing the result of Sayin et al.[2].

III. SIMULATION

We can now attempt to confirm the calculations above by developing a simulation. For the purposes of this exercise the problem has simplified in a number of ways. The two antennas are assumed to be isotropic emitters. This eliminates the need to adjust the antennas to track the satellite and in this imaging situation by restricting the possible target space, we can eliminate possible ambiguities caused by this simplification. The total number of pulses and therefore recording time will be kept to a value such that the target will remain in the 3 dB area of the directional antenna. This means that the output between the isotropic antennas and the directional ones will differ by less than 3 dB. The path of the satellite will be assumed to be straight eliminating complications caused by the changes in the geometry. Compensation for the curved path of the orbit is an exercise outside of the scope of this paper. A constant RCS is also assumed for all angles to reduce computation time.

A. Signal Reconstruction

In order to properly model the system described above, a model of the downlink signal must be synthesized in accordance with the findings of Humphreys et al. A MATLAB program was developed for this purpose. The output of this program is an array of complex values written to a MAT file. We begin by loading an arbitrary data file to be encoded. The program then determines the number

of frames required to encode the data. The program then begins creating the sections that do not change from frame to frame. It loads the known PSS hex values from a file before DPSK encoding them and repeating the sequence eight times. Note that this sequence is not OFDM and is rendered in the time domain. The sequence is then normalized and upsampled to match the other symbol lengths in time. The SSS sequence is then loaded and 4QAM modulated. The data is then multiplexed using the MATLAB OFDM method. The variable peak to average power ratio (PAPR) of OFDM encoding presents an issue to the average power when the signal is normalized. For this reason, the peaks of the output of the OFDM signal are clipped when they go above a certain threshold that can be set within the simulation. This issue and the resulting signal distortion caused by the clipping is discussed further in the results section. The data to be encoded in the header section is then read in from its file and encoded into three separate symbols, following the same procedure as the SSS. Because there is no known value for the CSS and CM1SS, a random sequence is generated for each and encoded similarly to the SSS.

The program then writes the encoded PSS, SSS, and header sections to the output file sequentially. The six symbols of payload data is then taken and encoded similarly to the SSS but using 16QAM instead of 4QAM. This is then written to the output. The remaining payload symbols from this frame are then encoded like SSS and written to the output. The CM1SS and CSS are written along with a guard interval of one frame of all zeros. This process is repeated frame by frame until all of the data file has been written into payloads. Also written into this program is the ability to change the sampling rate depending on if oversampling is needed.

B. Signal Propagation

The signal propagation program uses largely MATLAB tools to describe the evolution of the signal through out the environment. The raw waveform is loaded in from a file and then mixed with a carrier wave. The doppler shift from the movement of the satellite is here encoded into the signal. The simulation then has free space propagation path for each individual target and one direct path between the transmitter and the receiver. These paths model

the evolution of the wave as it travels through free space before hitting its designated target. This simulation can account for an unlimited number of the targets, however computation time and memory usage will increase as well. Next each channel is propagated to the receiver where they are combined and mixed down to an output signal which is then saved to a file. The target and direct path signal are saved to different files, so they can be compared in post processing. One of the side effects of this set up is that the direct path must be intentionally mixed with target signal. This allows the evaluation of direct path mitigation algorithms which is a topic for future work. This process is repeated for each pulse. The final output are two files representing the data cubes for both the target and reference signals.

C. Signal Processing

The signal processing program takes in the direct path and the target signal files, but does not load them into memory as these files can be very large (30+ Gb).

1) *Matched Filtering*: The first step in the processing chain is range compression using matched filtering. The cross correlation between the signal and the reference produces a peak at the difference in time of arrival.

2) *Back Projection*: For each pulse, the back projection algorithm calculates the distance to every point in the output frame. It then interpolates the range profile at each pixels range values and adds that complex value to the pixel. This is repeated for every pixel in every pulse. If the signals do originate from the same place, they coherently add at only the place where the target has reflected the signal. This is a textbook implementation of the back projection algorithm.[7]

IV. RESULTS

A. Waveform

We can now look at the waveform produced and compare them to the recorded values. Figure 8 shows clear similarity Figure 4 (c), except that its peak is clipped much more clearly. The clip ratio is a tuning variable in this simulation, and in this initial testing the value of $\sqrt{2}$ was used. The higher this peak values is allowed to climb, the lower the average power is when the power is normalized. The clipping algorithm implemented in

this simulation is fairly naive and better situations exist in the literature but are outside of the scope of this paper. It is clear that a different value is used in the actual transmissions. The gaps between frames can be seen, however the gaps in Figure 4 (b) are much bigger because of their incomplete frame occupancy. The power spectra (Figure 9) marks

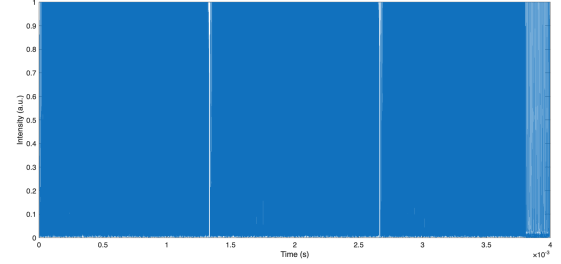


Fig. 8. Time series view of the three frames

a significant departure from what is seen in the literature. While a clear bandwidth can be made out, the frequency drop off outside of the 120 MHz is slow. This is an artifact of the clipping done to the OFDM signals. The clipping introduces sharp edges which introduce higher frequency components. This is a large problem in OFDM encoding and this distortion is balanced against the average power loss. For this simulation, the average power was prioritized to preserve SNR. We can see the clear peak expected from an autocorrelation of a non-repetitive signal. We can also see the clear peak at 1320 m corresponding to our cyclic prefix. There are also a large number of other lower peaks. There are several theories as to why these peaks appear. They could be related to the DPSK encoding and the repetition of that sequence causing peaks as has been discussed in Humpherys et al.[5]. This is an opportunity for future analysis, as there are no similar figures in literature.

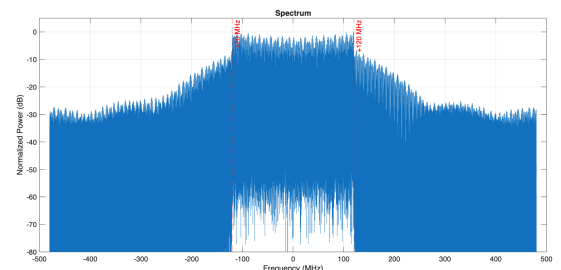


Fig. 9. Spectral view of the data at an increased sampling rate to show frequency fall off

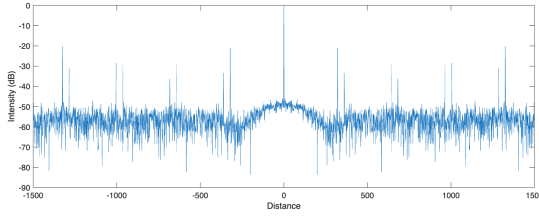
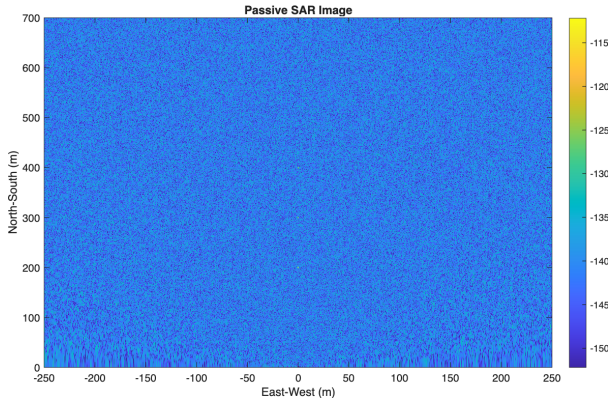


Fig. 10. Matched filter response of one frame.

B. SAR

Using our propagation and processing chain, we can now create a scene and simulate it. We place three targets on the X axis at 200 m, 300 m, and 400 m each with an RCS of 1000 m^2 . These are meant to symbolize targets on the order of a building similar to those seen in Gomez-del Hoyo [3]. The satellite begins its pass at -25000 m along the Y axis moving at 7.5 km/s in the positive Y, passing directly over the receiver. We then simulate 5000 pulses, as the satellite passes and coherently add our results. Because of the sharp peaks of our auto-

Fig. 11. SAR image of the three targets at 1000 m^2 RCS

correlation function, the target points in Figure 11 is almost indistinguishable from the background. Even in Figure 12 the distinction between the background and the target is hard to make out even though their values differ by 20 dB. We can now reduce the target RCS and see how our output changes once again. The RCS is now dropped to 100 m^2 for all the targets. The resolution of the back projection is decreased to decrease compute time however all other parameters are unchanged. We can see in Figure 13 and 14 that smaller targets on the order of vehicles are detectable in this regime, which is confidence building for future experiments. The theoretical max range resolution, calculated from

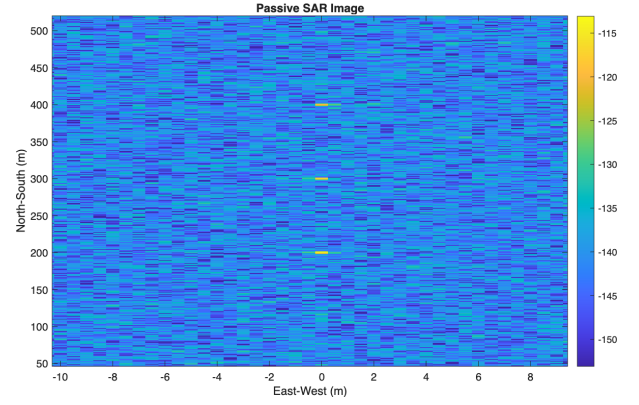
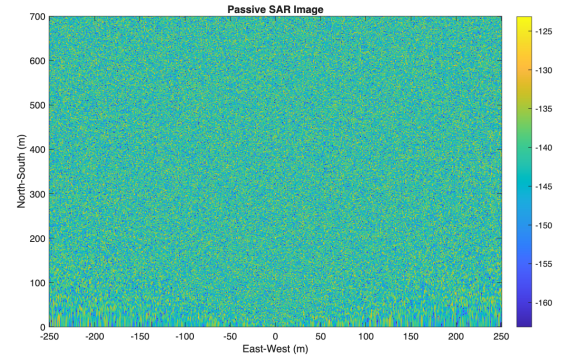


Fig. 12. Zoomed in version of Figure 11

Fig. 13. SAR image of three targets at 100 m^2 RCS

the bandwidth, is 0.6 m which is on par with the results seen in Figures 12 and 14.

V. CONCLUSION AND OUTLOOK

Overall, the future of passive imaging using the Starlink satellites is bright. It is still an active research field with many developments coming in the last few years. There were many future results alluded to in papers that have to be released. The work by Gomez-del Hoyo[3] shows that a IoO system for terrain mapping is possible. The simulation of that work was the original purpose of this project but complexity of the signal and simulation design required a change in the scope. The bistatic

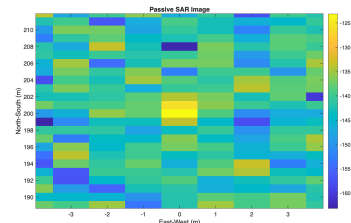


Fig. 14. Zoomed in version of one target in Figure 13

geometry and extreme distances mean that many existing MATLAB signal propagation frameworks do not work.

The system above is a useful framework for building understanding of how a passive imaging system using Starlink or any passive imaging system functions. It has many places for improvement. The effect of an angle variable RCS could be easily implemented without many changes or increased computation time. Compute time could be decreased by implementing parallelism, however this will still be limited by the total memory of the system. Implementing orbital geometry would not be difficult, but would require further processing work. The ultimate goal of terrain simulation would require a complete redesign of the system and greater thought devoted to run time. Hopefully, this system can be used as a tool in the future for further exploration into the world of passive radar.

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